THE UNIVERSITY OF SYDNEY

Major Assignment

AMME 2500

Contents

Abstract

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This report investigates the dynamic behaviour of a car's suspension system, and the effect that varying the spring and dampener coefficients has on the system's ability to absorb forces from traversing a bumpy road.

The system was analysed by first simplifying it, by modelling the car as a beam, and the wheels as points with a spring and a dampener each connected to the beam. Then Newtonian Mechanics were used to derive a coupled system of differential equations, and then solved numerically using the Runge-Kutta method.

The effects of varying the spring coefficient k and the dampening coefficient c were analysed. The system exhibited minimal vibrations for specific k and c values. This demonstrates that there are optimal parameters to make vehicles comfortable and smooth to drive, and modelling the system like this with specific parameters can be used as a starting point for suspension design.

1 Introduction

Dynamic systems with springs and dampeners are used widely in engineering applications. They are mainly used to react to and suppress external forces on a system to limit the effects that they pose. An example of this is a car's suspension system, which is used to minimise the vibrations of the chassis when on bumpy or rough surfaces. Modelling these dynamic systems is crucial to understand their behaviour under different conditions, which can then be used to optimise performance, stability and reliability.

In this report, the dynamics of a car's suspension system are modelled and analysed. The motivation behind this is because suspension systems are extremely important for ensuring a smooth driving experience. They consist of a spring and dampener system for each wheel, where

2 References